

Capacity Planning Example: URL Shortener

Worked Example with Formulas and Numbers

Assumptions:

- Total Users: 100M
- DAU: 10M (10% engagement)
- Actions per User per Day:
 - Redirects: 100
 - Shortens: 0.01 (~100K/day)
- Request size: Redirect log = 200 bytes

Step 1. Requests per Second (RPS)

Formula: $RPS = (DAU \times \text{Actions per User per Day}) / 86,400$

Redirects = $(10M \times 100) / 86,400 \approx 11,574$ RPS

Shortens = $100K / 86,400 \approx 1.15$ RPS

Step 2. Throughput per Second

Formula: $\text{Throughput (bytes/sec)} = RPS \times \text{Avg Request Size}$

Redirect logs = $11,574 \times 200B \approx 2.3$ MB/sec

Step 3. Daily Data Growth

Formula: $\text{Daily Growth} = \text{Throughput} \times 86,400$

= $2.3 \text{ MB/sec} \times 86,400 \approx 199$ GB/day

Step 4. Annual Data Growth

Formula: $\text{Annual Growth} = \text{Daily Growth} \times 365$

= $199 \text{ GB/day} \times 365 \approx 73$ TB/year

Step 5. Node Capacity Estimation

Formula: $\text{Nodes} = \text{Required RPS} \div \text{RPS per Node}$

Suppose 1 DB node can handle 2K writes/sec.

Required = $11,574 \div 2,000 \approx 6$ nodes (plus replicas).

Step 6. Peak Traffic Planning

Formula: $\text{Peak RPS} \approx 2-3 \times \text{Avg RPS}$

Peak RPS $\approx 3 \times 11,574 \approx 34,700$ RPS

Step 7. Cache Offload

Formula: $\text{Effective DB RPS} = \text{Total RPS} \times (1 - \text{Cache Hit Ratio})$

With 95% cache hit ratio: $11,574 \times 0.05 \approx 579$ RPS

Step 8. Network Bandwidth

Formula: $\text{Bandwidth (MB/sec)} = \text{Throughput} \div 1,048,576$

= $2.3 \text{ MB/sec} \div 1,048,576 \approx 0.0022$ GB/sec (~18.9 GB/day traffic)